The graphwise lattice-to-continuum lemma via a spectral regulator

TECT verification-first repository · claim A3-PERTURBATIVE-CONTINUUM-CORRELATORS

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1. Purpose and scope

This note proves the gap `A3-GRAPHWISE-CONVERGENCE` via a *spectral* (Galerkin) regulator, after the operator identified an aliasing flaw in the finite-difference-lattice proof of v1.1/v1.2 (§3). Scope: scalar, $d = 3$, $\mu^2 > 0$, $\gamma > 0$, perturbative. The result is the order-by-order continuum limit of the perturbative correlators under the spectral regularisation; the genuine-lattice version and regularisation-independence are the separate refinement of §8 (Route B). Tier stays T3 (operator's authorised tier); this is the corrected submission for T6 ratification. Ingredients: `codes/foundations/a3_graphwise_convergence_checks.py` (8/8 PASS, incl. the flaw demonstration and the spectral bound).

2. Setup and the spectral regulator

$F_{\text{TECT}} = \int[\frac{1}{2}\phi K(-i\nabla)\phi + \frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6]$, $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$, $G = 1/K$. Regularise by a sharp *spectral* cutoff on the continuum kernel,

$$G_a(q) := \frac{\mathbf{1}_{\{|q| \leq \pi/a\}}}{K(q)}, \qquad \Lambda_a := \pi/a, \tag{1}$$

i.e. a Galerkin truncation to modes $|q| \leq \Lambda_a$. A connected graph $\mathcal{G}$ (V vertices, I lines, L loops) has amplitude $\mathcal{A}_{\mathcal{G},a}(p) = \int_{(\mathbb{R}^3)^L} \prod_\ell \frac{d^3k_\ell}{(2\pi)^3} \prod_i G_a(q_i(k,p))$, q_i linear in (k,p).

2′. The measure is well-defined ($\gamma > 0$)

The correlators are expectations in $d\nu_{\Lambda,a} \propto e^{-F_{\Lambda,a}} d\phi$ on the finite-dimensional cutoff space. For $\gamma > 0$ the sextic dominates the quartic ($\frac{\gamma}{6}t^6 + \frac{\lambda}{4}t^4 \to +\infty$, any λ), so $F_{\Lambda,a}$ is bounded below and $Z_{\Lambda,a} < \infty$: the measure and its correlators are well-defined (hypothesis `A2-H2-SEXTIC-COERCIVITY`). The $\mathcal{A}_{\mathcal{G},a}$ are the coefficients of its λ, γ expansion.

3. The aliasing flaw of the finite-difference lattice (recorded)

With the finite-difference lattice momentum $\hat{q}_a = \frac{2}{a}\sin\frac{aq}{2}$, $\hat{q}_a$ is periodic. An internal-line momentum $q_i = k_1 + k_2 + p$ (sunset type) with $k_1, k_2 \in \text{BZ}$ can have $|q_i| \sim 2\pi/a$, where $\hat{q}_a(2\pi/a) = \frac{2}{a}\sin\pi = 0$ and $G_a^{\text{latt}}(q_i) = 1/(\mu^2 + Yq_0^4) = O(1)$ — NOT bounded by $C(1 + |q_i|)^{-4}$ (assert `lattice_folding_flaw_demonstrated`: the ratio $G_a^{\text{latt}}(1 + |q|)^4$ at $q \sim 2\pi/a$ blows up as $7.6 \times 10^7 \to 7.2 \times 10^{11}$). So the v1.1/v1.2 majorant fails on the Umklapp/aliasing region; the finite-difference lattice needs a genuine lattice power-counting theorem (§8, Route B). The spectral regulator (§2) avoids this because it carries the continuum kernel and a sharp cutoff on each line.

4. Pointwise convergence

For each fixed q, $\mathbf{1}_{\{|q|\le\pi/a\}} \to 1$ as $a \to 0$ (since $\Lambda_a \to \infty$), so $G_a(q) \to G(q)$ pointwise; the integrand of $\mathcal{A}_{\mathcal{G},a}$ converges pointwise a.e. to $\prod_i G(q_i)$.

5. Uniform domination (spectral)

Because $K(q) = \mu^2 + Y(q^2 - q_0^2)^2 \asymp (1+|q|)^4$ with a positive lower constant ($K(q) \ge c(1+|q|)^4$, $c = \inf_q K/(1+|q|)^4 > 0$), for *every* q

$$G_a(q) = \frac{\mathbf{1}_{\{|q|\le\pi/a\}}}{K(q)} \le \frac{1}{K(q)} \le \frac{C}{(1+|q|)^4}, \qquad C = \sup_q \frac{(1+|q|)^4}{K(q)} = 1.6\times 10^3, \tag{2}$$

with C *independent of* a (assert `spectral_regulator_uniform_bound`). The indicator only restricts the domain, so (2) holds for the *unwrapped* internal-line momenta q_i — exactly what failed for the lattice regulator. Hence the integrand is dominated by $\Phi(k,p) = \prod_{i=1}^{I} C(1+|q_i(k,p)|)^{-4}$, independent of a.

6. Dominated convergence and the single limit

By A3-UV every subgraph has Weinberg degree $D' = 3 - 3V' - I' < 0$, so $\Phi \in L^1((\mathbb{R}^3)^L)$ for compact p (assert `dominating_function_integrable_weinberg`). Dominated convergence gives, *for each fixed external momentum* p, $\lim_{a\to 0}\mathcal{A}_{\mathcal{G},a}(p) = \mathcal{A}_{\mathcal{G}}(p)$. (Uniform convergence on compact p-sets is not claimed: it would need an additional equicontinuity / uniform-tail estimate beyond the per-p majorant.) The cutoff is $\Lambda_a = \pi/a$, so $a \to 0 \equiv \Lambda \to \infty$ is a single limit.

Theorem (A3-GRAPHWISE-CONVERGENCE, spectral regulator). *For the scalar Brazovskii theory ($d = 3$, $\mu^2 > 0$, $\gamma > 0$) with the spectral regulator $G_a = \mathbf{1}_{\{|q|\le\pi/a\}}/K$, every connected perturbative amplitude satisfies, for each fixed external momentum p, $\lim_{a\to 0}\mathcal{A}_{\mathcal{G},a}(p) = \mathcal{A}_{\mathcal{G}}(p)$. Hence each order of the perturbative correlators has a cutoff-independent continuum limit.*

7. Devil's-advocate (three objections)

(α) *"The v1.1 lattice proof was wrong, so trust is low."* UPHELD and corrected: the aliasing flaw is recorded (§3) and the proof re-done with the spectral regulator, whose domination (2) provably holds for unwrapped momenta. The verification now explicitly exhibits the flaw and the fix.

(β) *"Spectral $\neq$ the finite-difference lattice of the numerics."* VALID with mitigation: this establishes the continuum limit under a legitimate (Galerkin) regularisation, which suffices for the perturbative continuum object. Regularisation-independence and the genuine lattice are the separate Route-B refinement (§8), not claimed here.

(γ) *"Order-by-order, not the full series."* VALID: perturbative only; resummation / the constructive limit is $\mathrm{A3}_{\text{constructive}}$.

Quantitative sanity check (mandatory): lattice folding ratio $G_a^{\text{latt}}(1+|q|)^4 \sim 10^7$–$10^{11}$ at $q \sim 2\pi/a$ (flaw); spectral constant $C = \sup_q (1+|q|)^4/K(q) = 1.6\times 10^3$, a-independent; pointwise $\mathbf{1} \to 1$; Weinberg $D = 3 - 3V - I < 0$; $\Lambda = \pi/a \to \infty$ (8/8 checks).

8. Route B (separate refinement): the genuine lattice

To prove the same convergence for the finite-difference lattice $\hat{q}_a$ (and thereby regularisation-independence) one needs a lattice power-counting theorem (Reisz) handling folded line momenta and Umklapp sectors: a uniform lattice majorant replacing (2), with the lattice degree of divergence controlled over the periodic BZ. This is a strictly harder, separate lemma, left open as a refinement; the perturbative continuum limit itself is already established by Route A.

9. Result footer

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Result ID:                 A3-PERTURBATIVE-CONTINUUM-CORRELATORS
Precise statement:         Under the SPECTRAL regulator G_a=1_{|q|<=pi/a}/K,
                           every connected perturbative amplitude converges
                           lim_{a->0} A_{G,a}(p)=A_G(p) for each FIXED p;
                           hence the perturbative correlators have a cutoff-
                           independent continuum limit. (gamma>0 measure.)
Scope:                     scalar, d=3, mu^2>0, gamma>0, perturbative;
                           spectral/Galerkin regulator (Route A).
Dependencies:              A1-KERNEL-CONV; A3-UV-SUPERRENORMALISABILITY.
Method:                    spectral cutoff + dominated convergence.
Open refinement:           genuine finite-difference lattice (Reisz power
                           counting; folded momenta / Umklapp) -- Route B.
Evidence grade:            PROVED CONDITIONAL (analytic) + EXECUTED (8/8).
Reproduction command:      python codes/foundations/
                           a3_graphwise_convergence_checks.py
Expected output:           8/8 PASS (flaw demonstrated + spectral bound).
Falsification gate:        a connected graph G with A_{G,a} not-> A_G under the
                           spectral regulator; or sup_q (1+|q|)^4/K(q)=inf.
Tier before / after:       T3 (operator-authorised) / re-submitted for T6.
No-overclaim statement:    Spectral regulator only (NOT the finite-difference
                           lattice -- that is Route B); order-by-order; NOT
                           resummation/constructive; not Class-II; not mu^2<0.
Next required action:      Operator ratification (spectral T6) and/or Route B
                           (Reisz lattice power counting) for the genuine lattice.
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